From: M. E.
To:
Subject: [External] Fwd: DOGE: Apple To Invest $500-Billion in AI-Data Center (Texas); How About AI-Centers of Excellence-COE?
Date: Tuesday, February 25, 2025 9:19:15 AM
Attachments:

Dear Professors and Associate Professors, University of Virginia,

I see that the future will require AI integration for all of America's Energy (Smart Grid) Infrastructure; AI for FAA's National Airspace Modernization and Advance Air Mobility-AAM; AI for Veteran Administration's-VA TeleMedicine; and many other applications.

Please see my company's below discussion with the General Services Administration-GSA pertaining to our Proposal to assume their Property Lease on the now-defunct Federal Executive Institute-FEI in Charlottesville, VA.
You can tell from our pitch that our company aims to repurpose this FEI Institute Campus into a new cluster of Artificial Intelligence-AI Technology Centers of Excellence-COE...with purpose of attracting R&D and technology innovators to the region to help us advance the proliferation of AI into various Sectors. We will then leverage our AI R&D activities towards the $-Hundreds of Billions being invested into AI-Data Centers in Virginia and around the USA. We will then push our AI expertise towards Europe.

After you speak with your UV-Dean on this matter, I hope to hear if University of Virginia-UV might be interested in being a collaborative part of our AI Technology Center of Excellence_COE plans, and what you feel that the UV-technology departments might be interested in contributing? What goals and objectives would UV desire as outcomes for your collaboration?

Respectfully,

Michael J. Erickson, USMC-Retired
President, LilyPad Airways, LLC

subject: **GSA, DOE, NREL, USDOT / FAA; DOS: Federal Executive Institute-FEI Property Development ?**

Dear Ms. Nicole and GSA,

As you now know...the Federal Executive Institute-FEI is an executive and management development and training center for governmental leaders located on a 14-acre campus near downtown Charlottesville, Virginia, less than a mile from the University of Virginia (Address: 1301 Emmet St N, Charlottesville, VA 22903). Because of its close proximity to University of Virginia...it might be very convenient for any of my below COE concepts, if we linked our new FEI campus activities to the University. For background, please see attached document and:

- https://www.opm.gov/services-for-agencies/center-for-leadership-development/federal-executive-institute/
- https://www.gsa.gov/real-estate/historic-preservation/explore-historic-buildings/find-a-building/federal-executive-institute-charlottesville-va

The FEI was historically open to career SES, Senior Level (SL), Scientific and Professional (ST), and SES Equivalent (e.g., under Title 10, Title 38, etc.) members from across Federal Government.

In light of President Trump's Executive Order-EO on Monday, which calls for the Elimination of the (GSA-maintained) Federal Executive Institute-FEI in Charlottesville, VA...

- https://www.whitehouse.gov/presidential-actions/2025/02/eliminating-the-federal-executive-institute/

As noted in my previous correspondences to you, we recognize GSA efforts to comply with President Trump's requirement to close the Federal Executive Institute-FEI in Charlottesville, VA.

We also understand that normally, when GSA gives up a property under your control, that other Federal Agencies might have opportunity to submit their Proposals for the property, before it is released upon the public.

One of your GSA-FAS senior managers within the **Travel, Transportation, & Logistics Division**...and who is currently one of the leaders in the **GSA Fleet Vehicle Management** program...is a very close personal friend of mine.

I understand that he accepted GSA's 'buyout' and is in process of retiring/leaving the GSA as a result of your planned downsizing of that Division.

As soon as he is free from GSA direct employment, we will offer him a position with our company leading our company's AI-Fleet Vehicle Management Technology Center of Excellence-COE activities...when it gets underway.

We are also offering to buy all of the GSA Fleet Vehicle Management pieces that you and GSA decides to 'Privatize'.

- We will then begin introducing our concepts for FedRAMP-compliant, and CyberSecure Artificial Intelligence-AI into how the nation and Federal Agencies operates Fleet Vehicle TeleMatics; Automated Vehicles; 'Smart Roadways'; Intelligent Transportation; etc. (https://www.gsa.gov/technology/government-it-initiatives/artificial-intelligence/ai-guidance-and-resources).
- We will collaborate with Federal Fleet Service Representatives across America (https://www.gsa.gov/buy-through-us/products-and-services/transportation-and-logistics-services/fleet-management/vehicle-leasing/find-a-fleet-service-representative)
- We will propose our own Technology Transfer (T2) Plan (https://www.transportation.gov/research-and-technology/technology-transfer)
- We will invite David Wollman's NIST Smart Connected Systems Division to review our 'Smart Transportation' & TeleMatics recommendations (https://www.nist.gov/programs-projects/nist-automated-vehicles-program).
- We will apply for DOT SMART Grants (https://www.transportation.gov/grants/SMART) (https://www.transportation.gov/policy/OST-R/grants).
- We will then invite Gregg Fleming's VOLPE, and Brian Cronin's DOT-ITS to validate our concepts. We anticipate needing the support of some other (former) GSA experts that may have also accepted the 'buyout'. Thus, we may be able to provide a 'softer-landing' for some of those personnel that you cull from GSA employment.

THUS, WE NEED THE COMBINED WILL OF SEVERAL BRANCHES OF FEDERAL GOVT AS WELL AS DOD / MILITARY ON THIS.

Our team feels that we have a very viable and productive strategy for repurposing the FEI Institute property, that helps GSA meets its Property needs, as well as many of the Federal Agencies to meet their AI & Aviation Technology needs...while still helping the US Taxpayers to save Govt Budget money.

We propose that the FEI campus will be converted via Public-Private Partnership (PPP) (or sold to us directly) and that the Campus be repurposed as a multi-sector Technology Centers of Excellence-COE:

- **Sector-1 will be devoted to Artificial Intelligence-AI Technology Center: The Objectives of this AI Technology COE:**
    - With funding Grant from President's OSTP; DOD; Natl Science Foundation-NSF; State Dept (Tech. Division); Natl Endowment For The Humanities; etc.
    - Focus to include:
        - Of particular importance is the investment in the development of assurance and trust in AI systems. Research in these areas is essential for using AI in all fields, but it is particularly important in systems that involve safety or applications in which AI decisions affect individuals, groups, communities, and the environment. Most AI R&D thus far has focused on the advancement of AI for individual tasks. Additional work is needed to solve increasingly difficult science and technology challenges covering multiple domains and applications, moving toward the vision of general-purpose AI. AI R&D increasingly attempts to consider how various areas of AI work can fit together into an integrated system.
        - Develop foundation models that are trained on large amounts of unlabeled data, usually using self-supervised learning, and can be adapted to many application

domains such as law, healthcare, and science. Innovations continue to advance the frontiers of what foundation models can do on language and image tasks. Familiar examples of large pretrained language models include BERT (Bidirectional Encoder Representations from Transformers), GPT-4 (Generative Pre-trained Transformer), and other AI systems with skills that might begin to resemble intelligence within certain domains. Additional R&D is necessary to minimize unwanted fabrications and harmful biases in generative AI.

- Conduct research to enhance the validity and reliability as well as security and resilience of these large models, especially in response to adversarial attacks. Further research is also needed to develop techniques for explaining and interpreting model outputs. Additional work is needed to address privacy concerns related to training models on such large corpuses of data. Finally, appropriate safeguards will need to be conceptualized and designed into these systems.
- Developing AI Systems and Simulations Across Real and Virtual Environments
- Enhancing the Perceptual Capabilities of AI Systems
- Developing More Capable and Reliable Robots
- Advancing Hardware for Improved AI
- Creating AI for Improved Hardware
- Embracing Sustainable AI and Computing Systems
- Develop Effective Methods for Human-AI Collaboration
- Seeking Improved Models and Metrics of Performance
- Cultivating Trust in Human-AI Interactions
- Pursuing Greater Understanding of Human-AI Systems
- Developing New Paradigms for AI Interactions and Collaborations
- Understand and Address the Ethical, Legal, and Societal Implications of AI
- Ensure the Safety and Security of AI Systems
- Develop Shared Public Datasets and Environments for AI Training and Testing
- Measure and Evaluate AI Systems through Standards and Benchmarks
- Better Understand the National AI R&D Workforce Needs
- Expand Public-Private Partnerships to Accelerate Advances in AI
- Establish a Principled and Coordinated Approach to International Collaboration in AI Research
- Evaluating Federal Agencies' Implementation of the NAIIA and Strategic Plan
- AI Data Centers & AI-Enabled Edge Data Centers, Energized by AI-Enabled Smart DERs & MicroGrid Energy Infrastructure
- Although President Trump and his fellow Billionaires are planning to build $500-Billion AI-Data Centers in Texas, Arizona, and other SW-area locations...we here in Northern Virginia need to convert our Data Centers towards AI-Capability also. Further, we need to Design-Build Distributed Energy-DERs to energize their infrastructure. That means we need Community-based (Campus) Smart MicroGrids, Battery Energy Storage-BESS; Grid-Integrated Smart Buildings-GEBs; and Small Modular (Nuclear) Reactors-SMRs here in Northern Virginia.

Loudoun County is reporting that they have an additional 117 data centers in the pipeline -- on top of the 199 data centers they already have on the ground. That will give Loudon County at least 316-Data Centers in one county. The land value of the Loudoun data centers reportedly surged by $19 billion -- a 78 percent increase over last year. The rapid growth in data centers has led the Virginia General Assembly to pass regulations on new data centers requiring extra local approval and ensuring placement is not too close to schools or residential housing. Dominion Energy Virginia has asked for an additional $900-Million in Ratepayer Money just to continue their Wind Farm boondoggle...and Virginia's feckless FCC is going to grant it to them.

- Investigations on AI Data Center Vendor progress (e.g. McKinsey Transformation, ABBYY, GBTEC, Chazey Partners, PRIME BPM, SS&C Blue Prism, Celonis, mindzie and Skan AI).

- **We see AI being particularly useful for Agencies & Business Sectors such as:**
  - **Veterans Admin-VA:**
    - AI-Enabled TeleMedicine;
    - AI-Enabled Privatization of VA's Claims Processing;
    - AI-Enabled Data Science as a Service (DSAAS)
    - *(We should contact VA's C.O. (Michael Berberich) for this. We would need to compete against COFORMA Partners for this.)*
  - **NIH:** AI-Enabled TeleMedicine;
    - Agentic AI For Healthcare (NIH-ARPA Solicitation 75N99225RFI106).
    - AI-driven Food Extraction: aid in ration development, manage macro/micronutrient content, and analyze sensory and field test and evaluation data. The goal is to improve data analysis, so that products meet their specific nutritional needs.
    - *We could invite UC-Davis or other Hospital to collaborate with us on this.*
  - AI-Enabled Hospital & Private Consumer TeleMedicine;
  - AI-enabled Sports & Fitness; Event Analysis
    - AI Fitness Apps
    - Data Privacy Consultations
    - Immersive VR & AR Workouts
    - Smart Home Gyms; Smart Mirrors; *(we would need to compete against Apple; Tonal; Tempo Studio; JRNY App; Future; and Peloton systems*
  - **Bureau of Indian Affairs (BIA):** AI-Tools: a robust AI tool that will initially serve one bureau, processing approximately 2,600 actions per year but must be capable of scaling to accommodate over 20,000 actions. The tool should be designed to facilitate the creation of essential pre-solicitation documents, including Statements of Work (SOWs), Independent Government Estimates (IGEs), Market Research Reports (MRRs), Acquisition Plans (APs), and related documentation.
  - **DOD;**
    - USAF ADVANA Program: $15-Billion Chief Digital and Artificial Intelligence Office (CDAO), Advana Program has a requirement for Advancing Artificial Intelligence Multiple Award Contract (AAMAC). *(We would need to compete against Booz-Allen for this)*
    - AI-Enabled Soldier Field Tactical Body Vital-Sign Monitoring (Apparel-woven Sensors; Smart Clothing).
    - AI-based Military Logistics & Supply Chain Management;
    - Information Extraction from scientific documents
    - AI-based Contract Fraud Detection;
    - Defense Logistics Agency-DLA: Smart (Intelligent) Inventory Forecaster: This AI solution predicts inventory needs with pinpoint accuracy, cutting waste and boosting profitability. Perfect for e-commerce and retail brands scaling fast.
  - **DOE & Energy Sector:** AI-enabled Community-based Smart MicroGrids Infra, BESS Batteries and SMR Reactors & Sim Reactors;
  - **AI-enabled Data Centers:** This is useful for President Trump's $500-Billion in AI Data Centers planned.
  - AI-enabled **Air Traffic Management** (including UAS-Drones & eVTOL)

and Airspace Management for DOT & FAA; (This could have prevented the Reagan Airport crash)

- Digitized Ships Registry (e.g. State Dept);
    - AI-Enabled File & Document Manager
- AI-enabled **External Revenue Service-ERS** (Tariff & Trade Collections);
    - Tariff enforcement is generally the province of U.S. Customs and Border Protection (CBP), which falls under the Department of Homeland Security (DHS). However, they will be preoccupied by Deportation of Illegal Aliens. ***Thus, we might find a niche in supporting AI-Enabled State Dept. or privatized ERS provider to gather these Tariffs.***
    - CBP's internal guidance requires it to refer any possible criminal violation to the relevant U.S. Attorney's Office for further investigation and possible prosecution, with no requirement of notice to the alleged offender. Beyond this, the Disruptive Technology Strike Force (DISTECH), a joint venture that was created in February 2023 to prevent the proliferation and illegal acquisition of sensitive technologies by nation-state adversaries, can quickly be repurposed for tariff enforcement. Active in more than 10 major U.S. cities, DISTECH combines the resources and personnel of DHS, the Federal Bureau of Investigation (FBI) and the Department of Commerce (DOC) with federal prosecutors to investigate and prosecute sanctions evasion and export control cases.
- AI-Enabled Vehicle Management
- GSA's $40M **Natl Capital Smart (IoT) Buildings** Contract
    - AI-Enabled Smart Home Management
- AI-enabled Non-Fungible Tokens (NFTs) for Patenting & Watermarking; AI-Logo Maker. ***We should contact USPTO's Whitney Lee for this since they have upcoming $12M Solicitation for it; we would need to compete against Clarivate Analytics for this).***
    - **(these could be a new lines of business for our colleague...Ola Para's...graphic arts help)**
- AI-enabled CryptoCurrency;
- AI-Powered Customer Persona Builder: No more guessing games. This AI tool analyzes your customer data to create hyper-detailed, actionable personas in seconds
- AI-powered Website Makeovers
- AI-Enabled Innovation Hub Content Creation
- AI-Based ChatBot for Business
- AI-driven Financial Portfolio Management ***(this could be a new line of business we sell to our colleague...Robb Mitchell's...financial management company)***
- AI-Marketing Agency (we could start our own new Marketing Strategy Startup & Consultation; AI-Advertising Software); AI Presentation Generator: Need an investor deck or killer pitch? This AI builds stunning, data-backed presentations in minutes – just upload your data and watch it work.
- AI-Personalized Shopping Experience for Customers at Department Stores and Malls;
    - AI-enabled Checkout & Frictionless Customer Experience;
    - Enhanced Security & Loss Protection (Shoplifters);
    - Omni-Channel Integration & Seamless Shopping Journeys;
    - Dynamic Pricing & Personalized Offers / Discounts
    - Retail Customer Insights & Consumer Behavior Analysis

Enhanced Customer Service

- Hyper-Localized AI Ads: Why target broad demographics when this AI crafts ad copy tailored to micro-audiences based on real-time trends and regional insights? Say hello to record-breaking ROI.
- AI Video Editor on Demand: Editing. This AI tool creates pro-level videos from raw footage – with motion graphics, transitions, and effects – all in a fraction of the time.
- AI-enabled Online Educational Platforms; use of Synthetic Data for Training;
  - Perform data collection and research that supports strategic human capital initiatives, to include but not limited to: strategic human resources management; talent management; employee engagement; human capital analytics; diversity, equity, inclusion, and accessibility; sustainability; finance; business operations; digital transformation; cybersecurity; change management; data literacy; talent development; strategic recruitment; employee benefits and compensation; labor and employee relations; employee retention; upskilling and reskilling leaders and employees; improving overall culture and attitudes; and innovation to include artificial intelligence. This data and research must support the delivery of leadership development opportunities and human capital topics via asynchronous and synchronous delivery.
- AI-enabled (digital) Concierge for Hotels & businesses) Service
- AI-enabled Smart Agriculture solutions;
- AI-enabled Business Startup Consulting:
  - Business Modeling; Competitive Data Search;
  - Brainstorm to Launch: 6 Simple Steps

    1. Validate Your AI Vision:
    2. Prototype & Refine:
    3. Build Your MVP: No need for extensive coding knowledge! Embrace the power of no-code platforms like Bubble and RapidAPI to construct your Minimum Viable Product (MVP). An MVP should focus on core features that deliver your AI solution's value proposition. Start small, test its effectiveness, and gather valuable user data before expanding.
    4. Prepare for Liftoff:
    While your app takes flight, build anticipation. Create social media accounts, develop engaging content, and generate buzz around your AI-powered offering. This pre-launch phase is crucial for attracting early adopters and building excitement for your official debut.
    5. Launch & Take Flight:
    It's launch day! Ensure your landing page, demo video, and sign-up process are polished and persuasive. Consider platforms like Product Hunt to gain valuable exposure and attract your first wave of users. Remember, launching is just the beginning - continuous improvement and user feedback are key to sustained success.
    6. Grow Organically:
      - AI Lead Generation & Engagement;

- **Sector-2 will be devoted to AI-Enabled Advanced Air Mobility-AAM Technologies Center**
  - Funding Grant from DOT / FAA; NASA-UTM; VOLPE; etc.
  - Focus to include:
    - National Airspace System-NAS AI-Enabled Airspace Modernization:
      - Unmanned Aerial Systems-UAS (Drones);
      - eVTOL Passenger Air Transport

- Update FAA's Roadmap for Artificial Intelligence Safety Assurance

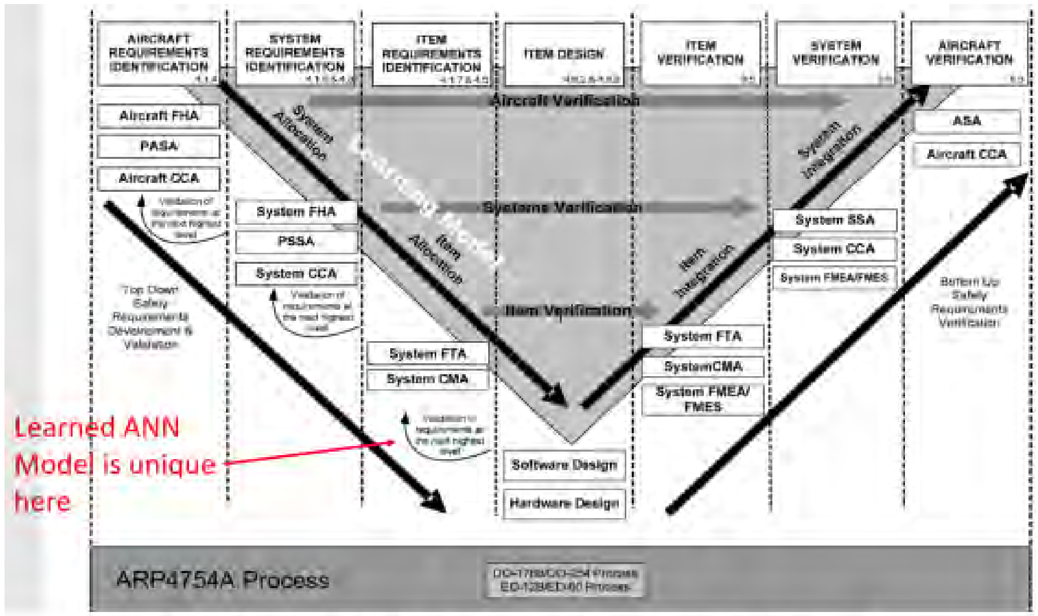

Figure 2 Development of AI Algorithm May Require Additional Adaptation (at Item Level)

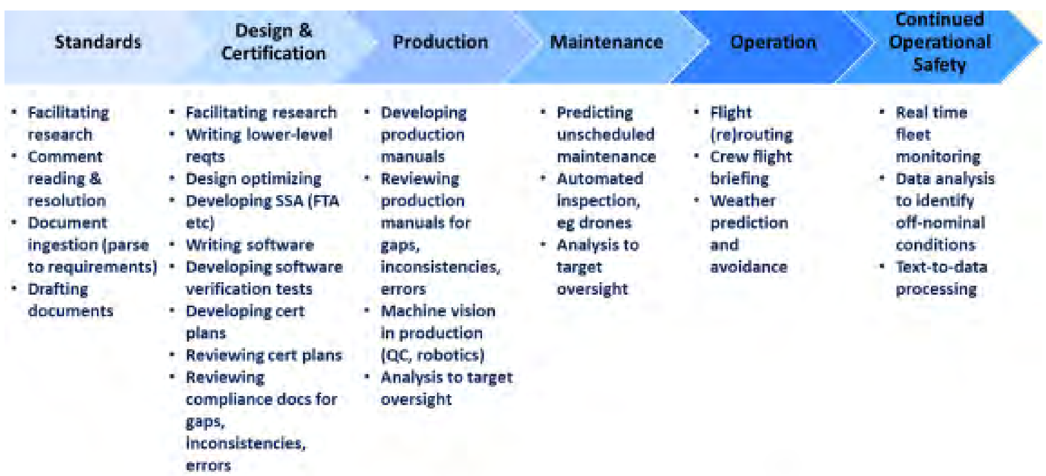

Figure 3 Example Use Cases for AI in Aircraft Safety Lifecycle

- **Sector-3 will be devoted to AI-Enabled Distributed Energy-DERs Technologies Center**
  - Funding Grant from NREL; Dept. of Energy-DOE, Office of Technology Transitions (see: https://www.connectwerx.org/portfolio-items/ppo-cwx-010-gdo-accelerating-interconnection-through-ai-ai4ax/)
  - Focus to include:
    - Campus Smart MicroGrid
    - Campus Battery Energy Storage-BESS
    - Campus-wide Grid Integrated Efficient (Smart) Buildings-GEBs
    - Campus Virtual Power Plant (VPP) &/or Small Modular (Nuclear) Reactor (SMR)
    - Develop partnerships between software developers, grid operators -- including Regional Transmission Operators and Power Marketing Administrations -- and energy project developers “to modernize the interconnection application

process and significantly reduce the time required to review, approve, and commission new generation interconnections across the country.
  - Optimize the placement of DERS by analyzing data on the location of DERs, the load profile of the grid and the weather forecast.
  - Control DERs in real time to optimize their operation and to ensure they are working in harmony with the grid.
  - Manage demand for electricity by incentivizing consumers to reduce their energy use during peak demand times.
    - Data collection and integration
    - AI algorithm selection
    - Load forecasting
    - DER optimization
    - Demand response management: Implement AI systems that adjust energy consumption in response to signals from the grid or market conditions. This can involve incentivizing users to shift their usage to off-peak hours.

- **Sector-4 will be devoted to AI-Enabled Fleet Vehicle TeleMatics Technologies Center**
  - Funding Grant from USDOT; Virginia DOT; FTA; GSA-EVSE Program Office; ARPA-I; etc.
  - GSA-FAS needs to sell off their bloated Federal Fleet Vehicle Management program and down-size that entire division. Please take a hard look at that GSA business unit. It is vital that the U.S. Federal Vehicle Fleet nationwide should be inculcated with Artificial Intelligence-AI throughout its network and Vehicle Management, as well as its Vehicle TeleMatics.
  - GSA will be forced to downsize, which will likely require GSA to sell off most of the Federal Vehicle Fleet Management business unit.
    Our Team desires to buy the entire GSA Fleet Management Division...and relocate it down to Charlottesville, VA so that we can integrate AI-technologies into the GSA Vehicle network.
    If GSA will give us the FEI Institute campus, then we will be able to convince someone like Enterprise Vehicle Leasing, or HERTZ Leasing, to join with us for purchase of the GSA Fleet Management Division.
  - FedRAMP & CyberSecurity Compliance
  - Safe and Responsible Use of Artificial Intelligence in Transportation
    - https://www.transportation.gov/sites/dot.gov/files/2024-12/ARPA-I%20Summary%20Report%20on%20AI%20RFI%202024-12%20%28Final%29_1.pdf
    - wide variety of areas broadly defined as asset management with computer vision, AI assurance, logistics, planning, project delivery, safety, accessibility, automated vehicles, as well as potential cybersecurity advancements. Improvements in aviation safety and efficiency were two other areas of possible progress mentioned by respondents.
    - Increased use of digital twins to simulate various aspects of transportation such as traffic, roadway structures, pedestrians, and human behaviors. Digital twins allow for testing new technology and ideas without putting humans at risk. Respondents noted that increasing the efficiency of large data storage collation to accurately predict and understand near-miss incidents may lead to important safety developments across several modes of transportation. Recommendations for future exploration that ARPA-I could support included the utilization of AI to better predict and understand traffic flows to better optimize traffic on our roads.
  - Transit Service with Smart Dispatch using AI technologies
  - Automated Vehicles & Driving Policy Within a 'Smart City' program, needs on Data, First Responders, and the Workforce

(see: https://www.transportation.gov/sites/dot.gov/files/2021-01/USDOT_AVCP.pdf)
    - The Role of Technology in Improving Project Delivery
    - Emerging, Overlooked, and Underleveraged Innovation for Safety

- **Sector-5 will be devoted to AI-Enabled World Diplomacy Technologies Center**
    - Funding Grant from State Department
    - Modernizing U.S. Diplomatic Statecraft through AI use & initiatives
        - https://www.state.gov/artificial-intelligence
        - https://2021-2025.state.gov/department-of-state-ai-inventory-2024/
    - Creation of new AI marketplace to bring specialized AI capabilities to a myriad of State employees.
    - From citizen services to foreign policy analyses and even negotiation advantages, AI offers the State Department opportunities to enhance Department efforts with original insights and beyond-human processing speed.
    - Investigate and advance scientific and technological capabilities and promote democracy and human rights by working together to identify and seize the opportunities while meeting the challenges by promoting shared norms and agreements on the responsible use of AI.
    - Together with our allies and partners, the Department of State promotes an international policy environment and works to build partnerships that further our capabilities in AI technologies, protect our national and economic security, and promote our values. Accordingly, the Department engages in various bilateral and multilateral discussions to support responsible development, deployment, use, and governance of trustworthy AI technologies.

Please direct me to the correct GSA Executive that can discuss repurposing the FEI Institute as our envisioned Technology Centers of Excellence-COEs...either by selling it to us, or within a Public-Private Partnership (PPP) type of arrangement, wherein we assume the FEI Property Lease.

Regards,

Michael J. Erickson, USMC-Retired
President, LilyPad Airways, LLC